REB, The Book
Example 6.6.1 Solution

The amount of A, (10%), in a gas mixture with B (65%), and inert, I, (25%) at 165 °C and 5 atm needs to be reduced to 0.1%. This is going to be accomplished by converting the A into Z according to reaction (1). An adiabatic, steady-state CSTR is going to be used. The rate expression is given in equation (2) where the pre-exponential factor is 1.37 x 10^5 m^3 mol^{-1} min^{-1} and the activation energy is 11,100 cal mol^{-1}. The heat of reaction is constant and equal to –7200 cal mol^{-1}, and the heat capacities of A, B, I, and Z are equal to 7.6, 8.2, 4.3, and 13.8 cal mol^{-1} K^{-1}, respectively. What space time is required, and what temperature will result?

$$A + B \rightarrow Z \quad (1)$$

$$r_1 = k_1 C_A C_B \quad (2)$$

Assignment Summary

Reaction

$$A + B \rightarrow Z \quad (1)$$

Rate Expressions

$$r_1 = k_1 C_A C_B \quad (2)$$

Reactor: Adiabatic, steady-state CSTR

Given Constants: $y_{A,in}$ = 0.1, $y_{B,in}$ = 0.65, $y_{I,in}$ 0.25, T_{in} = (165 +273.15) K, P = 5 atm, y_A = 0.001, $k_{0,1}$ = 1.37 x 10^5 m^3 mol^{-1} min^{-1}, E_1 = 11,100 cal mol^{-1}, ΔH_1 = –7200 cal mol^{-1}, $\hat{C}_{p,A}$ = 7.6 cal mol^{-1} K^{-1}, $\hat{C}_{p,B}$ = 8.2 cal mol^{-1} $^{-1}$, $\hat{C}_{p,I}$ = 4.3 cal mol^{-1} K^{-1}, and $\hat{C}_{p,Z}$ = 13.8 cal mol^{-1} K^{-1}.

Basis: $\dot{V}_{in}$ = 1.0 m^3 min^{-1}

Deliverables: τ and T

Completion Strategy

This assignment provides the desired response (outlet mole fraction of reagent A) from a steady-state CSTR. It asks for the corresponding temperature and the necessary space time. The reactor volume, outlet molar flow rates of the other reagents and outlet temperature can be found directly by solving the steady-state CSTR design equations. Having assumed the volumetric feed rate as a basis, the volume found by solving the design equations can then be used to calculate the space time.

Formulation of the Calculations

CSTR Model Function

CSTR Design Equations:

$$0 = \dot{n}_{A,in} - \dot{n}_A - Vr_1 = \epsilon_1 \tag{3}$$

$$0 = \dot{n}_{B,in} - \dot{n}_B - Vr_1 = \epsilon_2 \tag{4}$$

$$0 = \dot{n}_{I,in} - \dot{n}_I = \epsilon_3 \tag{5}$$

$$0 = -\dot{n}_Z + Vr_1 = \epsilon_4 \tag{6}$$

$$0 = -\left(\dot{n}_{A,in}\hat{C}_{p,A} + \dot{n}_{B,in}\hat{C}_{p,B} + \dot{n}_{I,in}\hat{C}_{p,I}\right)\left(T - T_{in}\right) - Vr_1\Delta H_1 = \epsilon_5 \tag{7}$$

CSTR Model Variables: V, $\dot{n}_B$, $\dot{n}_I$, $\dot{n}_Z$, and T

Additional Computable Unknowns: $\dot{n}_{A,in}$, $\dot{n}_A$, r_1, k_1, C_A, C_B, $\dot{n}_{B,in}$, and $\dot{n}_{I,in}$

$$\dot{n}_{i,in} = y_{i,in}\frac{P\dot{V}_{in}}{RT} \qquad i = A, B, \text{ and } I \tag{8}$$

$$y_A = \frac{\dot{n}_A}{\dot{n}_A + \dot{n}_B + \dot{n}_I + \dot{n}_Z} \qquad \Rightarrow \qquad \dot{n}_A = \frac{\left(\dot{n}_B + \dot{n}_I + \dot{n}_Z\right)y_A}{1 - y_A} \tag{9}$$

$$r_1 = k_1 C_A C_B \tag{2}$$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{10}$$

$$C_i = \frac{\dot{n}_i}{\dot{n}_A + \dot{n}_B + \dot{n}_I + \dot{n}_Z} \frac{P}{RT} \qquad i = A \text{ and } B \tag{11}$$

ATE Solver Requirements:

1. Initial guesses for the CSTR model variables (provided as an argument to the CSTR model function).
2. Residuals function that
   a. receives guesses for the CSTR model variables (V, $\dot{n}_B$, $\dot{n}_I$, $\dot{n}_Z$, and T)
   b. calculates the additional unknowns using equations (2) and (8) - (11)
   c. evaluates and returns the CSTR design equation residuals, equations (3) - (7).

Deliverables Function

1. Define an initial guess for the CSTR model variables.

$$V_{guess} = 1\ m^3 \tag{12}$$

$$\dot{n}_{i,guess} = \dot{n}_{i,in} \qquad i = B, I, \text{ and } Z \tag{13}$$

$$T_{guess} = T_{in} + 5\ K \tag{14}$$

2. Call the CSTR model function, passing the initial guess as an argument, to solve the design equations for the CSTR model variables.
3. Calculate the space time.

$$\tau = \frac{V}{\dot{V}_{in}} \tag{15}$$

Implementation of the Calculations

The Python script, example_6_6_1.py, and Matlab function, example_6_6_1.m, that were written to perform the calculations accompany this solution.

Results and Discussion

The ATE solver converged using the initial guess shown in equations (15) through (17). In this analysis, the initial composition and the conversion were specified. As such, the amount of heat released is fixed, and since the reactor is adiabatic, that also fixes the final temperature. Consequently, there is no need to check for other steady states.

The outlet temperature is 265 °C and the space time required to reach the specified outlet mole fraction of A is 0.42 min. The increase in temperature is expected because the reactor is adiabatic and the reaction is exothermic.

Summary

A space time of 0.42 min is required to reduce the amount of A to 0.1%. Doing so will increase the temperature of the gas mixture to 265 °C.